Journal of Machine Learning Research 0 (2026) 1-9 Submitted 5/26; Published

Detecting Timeout Effects in the NBA via the Generalized Likelihood Ratio Test

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Editor:

Abstract

We frame the question "does an NBA timeout move the game?" as a battery of one-sample Gaussian signal-detection problems. For three impact metrics derived from the cdnnba 2020–2025 play-by-play feed—excess net points, excess points-per-possession (PPP), and win-probability added—we test $H_0: \mu = 0$ against $H_1: \mu \neq 0$ via the generalized likelihood ratio statistic $G = -2 \log \Lambda$, which under H_0 is asymptotically χ^2_1. The three metrics are evaluated separately for coach full timeouts, coach challenges, and inferred mandatory television timeouts, and within each subtype we stratify by score margin, streak, game phase, home court, and regular season vs. playoffs. The point-differential metric collapses to a null result aggregated, but reveals a structured regression-to-mean signal once stratified by margin or streak. The win-probability metric rejects H_0 overwhelmingly: full timeouts cost the calling team roughly 3.9 percentage points of WP across a ± 6 possession window, challenges add roughly 1.6 pp, and television timeouts cost 1.3 pp—a pattern consistent with the prior causal literature. Across 163 stratified GLRTs, we reject H_0 in 69 cases (42.3%), versus the $\sim$5% expected if the null held everywhere.

Keywords: Generalized likelihood ratio test, signal detection, sports analytics, NBA, timeouts, win probability

1 Problem Statement

The NBA play-by-play feed records, for every coach-called timeout, the game state immediately before and after the stoppage. Whether timeouts actually "do something" to the game is contested in the literature (Assis et al., 2020; Gibbs et al., 2021; Weimer et al., 2023; Qiu et al., 2025). We treat each of three derived per-event impact metrics as a sample from a real-valued source and ask the binary signal-detection question "is the source mean zero?"

Concretely, let $X_1, \dots, X_n$ be the impact-metric values across timeouts within some stratum (e.g. full coach timeouts called while leading by 6–15 points), modeled as iid $\mathcal{N}(\mu, \sigma^2)$ with both parameters unknown. We test

$$H_0 : \mu = 0 \qquad \text{vs.} \qquad H_1 : \mu \neq 0. \tag{1}$$

A rejection means the timeout type, conditioned on the stratum, is associated with a non-zero shift in the metric.

2 Data and Metrics

We use the cdnnba enriched play-by-play feed for the 2020–2025 NBA seasons (regular season and playoffs), spanning 4,183,347 rows across 7,457 games and 1,480,697 possessions. Mandatory television timeouts are not labeled in the feed; we inject them via a deterministic rulebook procedure validated against the explicitly labeled 2013–2016 nbastatsv3 feed (F1 = 0.88–0.92). After injection, the cdnnba era contains 74,236 `full` coach timeouts, 1,396 `challenge` events, and 14,538 inferred `official_inferred` (TV) timeouts.

For each timeout event we look at a window of $W = 6$ possessions before and W after, and compute three metrics from the calling team's perspective.

Q1. `excess_net_post` Calling-team net points (for–against) in the post window, minus the season-PPP-gap expectation $(\text{PPP}_{\text{calling}} - \text{PPP}_{\text{opponent}}) \cdot W/2$. $\mu = 0$ corresponds to the calling team performing at the season expectation.

Q2. `excess_ppp_for_post` Calling-team PPP in the post window minus that team's full-season PPP. $\mu = 0$ means the team scores at season pace.

Q3. `wp_added_calling` Change in calling-team win probability across the $\pm W$ possession window, where Pr(home wins) is the output of a logistic-regression model trained on possession-start snapshots with the ten features score margin, period, seconds remaining, clutch indicator, possessing-team-is-home indicator, streak, playoff indicator, home win-percentage, away win-percentage, and the home-minus-away win-percentage difference. $\mu = 0$ means the window had no net WP movement for the calling team.

The WP model achieves AUC 0.844 and Brier score 0.161 on a held-out-by-game test set ($n = 59{,}860$).

For each metric we run the GLRT separately for the three timeout subtypes and stratify on five context columns from the project README: `is_home_calling`, `IsPlayoff`, `time_bucket` (Q1, Q2, Q3, Q4_early, Q4_clutch, OT), `margin_bucket` (seven buckets from down-15 to up-15 from the calling team's perspective), and `streak_bucket` (seven buckets from a 10-pt opposing run to a 10-pt own run).

3 Methods: GLRT Derivation

For $X_1, \ldots, X_n$ iid $\mathcal{N}(\mu, \sigma^2)$ the log-likelihood is

$$\log L(\mu, \sigma^2) = -\tfrac{n}{2} \log(2\pi\sigma^2) - \frac{1}{2\sigma^2} \sum_{i=1}^{n} (x_i - \mu)^2.$$

Under H_0 ($\mu = 0$) the MLE is $\hat{\sigma}_0^2 = \frac{1}{n} \sum x_i^2$; under H_1 the MLEs are $\hat{\mu}_1 = \bar{x}$ and $\hat{\sigma}_1^2 = \frac{1}{n} \sum (x_i - \bar{x})^2$. The supremum log-likelihood under each hypothesis evaluates to $-\frac{n}{2}(1 + \log(2\pi\hat{\sigma}_*^2))$, so the generalized likelihood ratio is

$$\Lambda \;=\; \frac{\sup_{\sigma^2} L(0, \sigma^2)}{\sup_{\mu,\sigma^2} L(\mu, \sigma^2)} \;=\; \left(\frac{\hat{\sigma}_1^2}{\hat{\sigma}_0^2}\right)^{n/2}. \tag{2}$$

Metric	Subtype	n	$\bar{x}$	G	p	sig.
Q1: excess net post	full	74,236	−0.011	0.98	0.32	ns
	challenge	1,396	−0.112	2.16	0.14	ns
	official_inferred	14,538	−0.003	0.01	0.90	ns
Q2: excess PPP for post	full	74,236	+0.007	7.12	0.008	**
	challenge	1,396	−0.021	1.42	0.23	ns
	official_inferred	14,538	+0.013	5.59	0.018	*
Q3: WP added (calling)	full	74,236	−0.039	10,881	$< 10^{-300}$	***
	challenge	1,396	+0.016	34.3	4.7×10^{-9}	***
	official_inferred	14,538	−0.013	245.2	$< 10^{-50}$	***

Table 1: Aggregated GLRT results by metric and timeout subtype. Significance: ns ($p \geq 0.05$), $*$ ($p < 0.05$), $**$ ($p < 0.01$), $***$ ($p < 0.001$).

Using the identity $\hat{\sigma}_0^2 = \hat{\sigma}_1^2 + \bar{x}^2$ we obtain the test statistic

$$G \;=\; -2\log\Lambda \;=\; n\,\log\Bigl(1+\frac{\bar{x}^2}{\hat{\sigma}_1^2}\Bigr), \tag{3}$$

which by Wilks' theorem is asymptotically χ_1^2 under H_0. We reject H_0 at level $\alpha = 0.05$ when $G > \chi^2_{1,0.95} \approx 3.841$. Equivalently, with $s^2 = \frac{n}{n-1}\hat{\sigma}_1^2$ and the one-sample t-statistic $t = \bar{x}/(s/\sqrt{n})$,

$$G = n\,\log\left(1+\frac{t^2}{n-1}\right) \xrightarrow{n\to\infty} t^2,$$

so the GLRT and the standard t-test are asymptotically equivalent; we report both throughout for cross-checking.

4 Results

4.1 Aggregated GLRT Results

Table 1 reports the GLRT statistic for each (metric, subtype) pair pooled over the entire cdnnba era.

Q1. Pooled over the league, the calling team's net post-window points cannot be statistically distinguished from the season-PPP expectation: $G = 0.98$, $p = 0.32$ for the most populous `full` subtype. This is the canonical null result reported by Assis et al. (2020) and is consistent with the regression-to-the-mean explanation: timeouts are most often called from a deficit, and the gap between pre-timeout net ($\bar{x}_{\text{pre}} = -1.73$ pts) and post-timeout net ($\bar{x}_{\text{post}} = -0.02$ pts) absorbs into the season baseline once we subtract the PPP gap.

Q2. The PPP metric—which removes the opponent's contribution—rejects H_0 for both `full` (+0.007 PPP, $p = 0.008$) and `official_inferred` (+0.013 PPP, $p = 0.018$). The effect is small but real: full coach timeouts add about 0.7% of a possession's worth of scoring relative to the calling team's season average, and TV timeouts add about 1.3%.

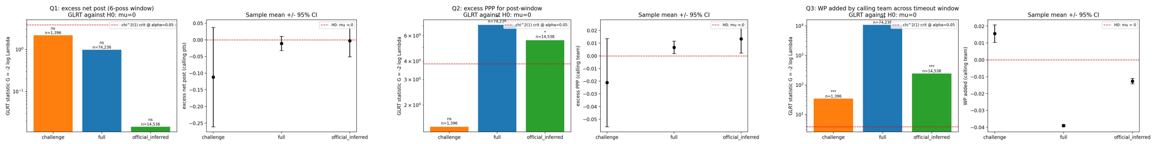

Figure 1: Aggregated GLRT statistics and means for Q1 (left), Q2 (middle), and Q3 (right). The dashed red line on the left panel of each plot is $\chi^2_{1,0.95} = 3.841$.

Q3. The WP metric overwhelmingly rejects H_0 for every subtype. Full timeouts cost the calling team an average of 3.9 WP percentage points across the ± 6 possession window ($G = 1.09 \times 10^4$), challenges *add* 1.6 pp ($G = 34$, $p < 10^{-8}$), and TV timeouts cost 1.3 pp ($G = 245$). The challenge sign is intuitive—challenges are issued when a team believes it has been wronged, so the post-window typically reflects a restored possession or extra free throws—but the negative WP for full coach timeouts is striking, and aligns with the slightly negative ATT that Gibbs et al. (2021) found via genetic matching.

Figure 1 shows the GLRT statistics on a log scale alongside the sample mean with 95% confidence intervals. The huge spread between Q1 and Q3 visualizes the value of measuring impact in WP units rather than in raw point swings.

4.2 Stratified by Pre-Timeout Streak

The streak bucket categorizes the calling team's signed streak (positive when the calling team itself has been on a run). Figure 2 shows that the WP signal is highly heterogeneous: timeouts called *into* a 6–9-point opposing run cost the full-timeout caller -10.3 pp of WP ($G = 12{,}992$, $n = 15{,}219$); timeouts called while the calling team itself is on a 6–9-point run *add* $+9.5$ pp ($G = 1{,}135$, $n = 1{,}470$). Crucially, the magnitude of the signed effect is roughly symmetric around streak $= 0$: timeouts do not appear to reverse momentum within the post window—they appear to ride it, regardless of which side the timeout was called from.

The same stratification on Q1 (Figure 3) is far less sharp—the signal is largely absorbed by the PPP-gap baseline—but the direction matches: down-streak timeouts produce small positive Q1 excesses, up-streak timeouts produce small negative ones.

4.3 Stratified by Score Margin

Stratifying on the calling team's score margin (Figure 4) exposes a very clean regression-to-mean pattern in the point-differential metric that is hidden by aggregation. For `full` timeouts, all three down-by buckets show positive Q1 excess (`down15+`: $+0.17$, $G = 24.4$; `down6-15`: $+0.11$, $G = 24.9$) while all three up-by buckets show negative Q1 excess (`up15+`: -0.26, $G = 39.5$; `up6-15`: -0.16, $G = 37.8$). The same monotone pattern holds, weaker, for inferred TV timeouts, but the challenge subtype shows no margin signal due to small sample sizes.

The Q3 margin stratification (Figure 5) tells a complementary story. Full-timeout WP loss is largest in moderately adverse situations (`down6-15`: -6.1 pp, $G = 6{,}749$; `down1-5`:

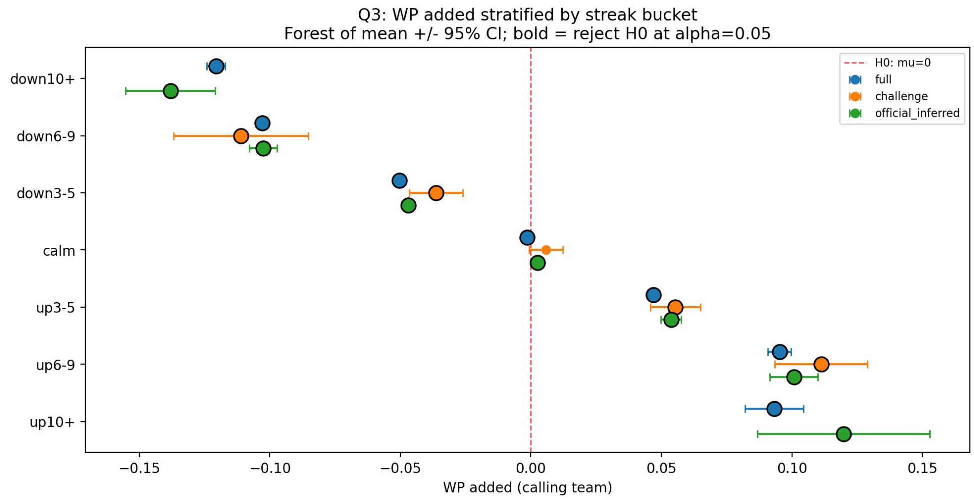

Figure 2: Q3 WP added stratified by pre-timeout streak. Bold filled markers indicate strata where $H_0\colon \mu = 0$ is rejected at $\alpha = 0.05$.

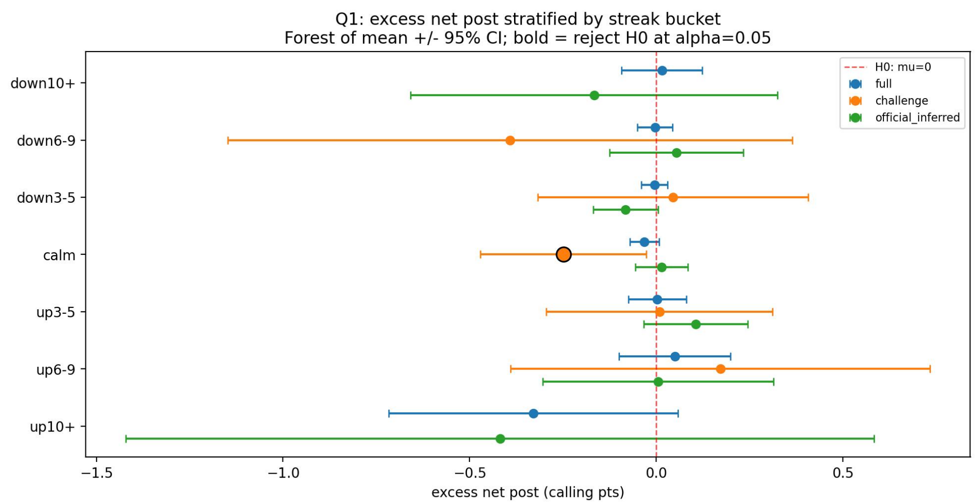

Figure 3: Q1 excess net post stratified by pre-timeout streak. Bold filled markers indicate strata where H_0 is rejected at $\alpha = 0.05$.

-5.3 pp, $G = 2{,}562$) and shrinks toward zero in extreme leads or deficits where the win is largely already decided.

4.4 Stratified by Game Phase

Figure 6 plots Q3 by quarter, with Q4 split into early (> 5 min remaining) and clutch (≤ 5 min). Full-timeout WP cost is roughly constant across Q1–Q3 (-4.0 to -4.5 pp) and OT

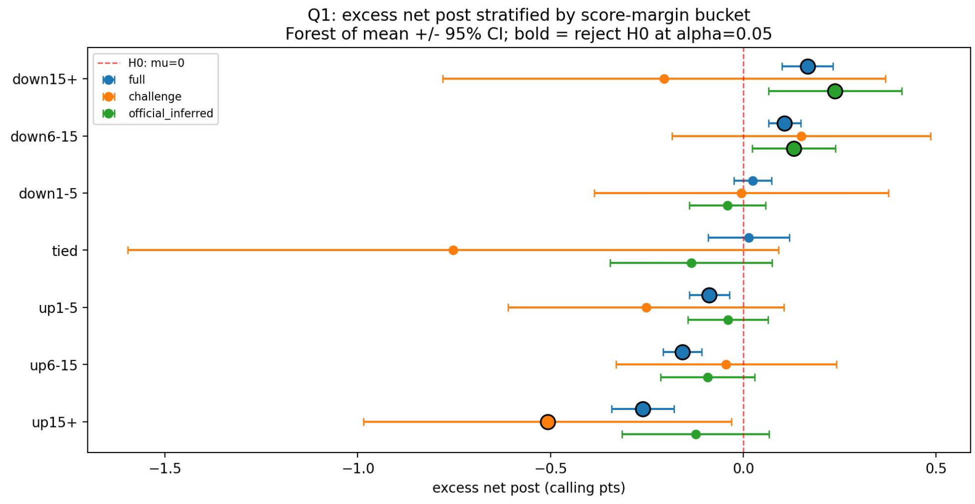

Figure 4: Q1 excess net post stratified by calling-team score margin. The left-to-right monotone trend is the regression-to-mean signature: trailing teams gain back roughly what they were missing, leading teams give some back.

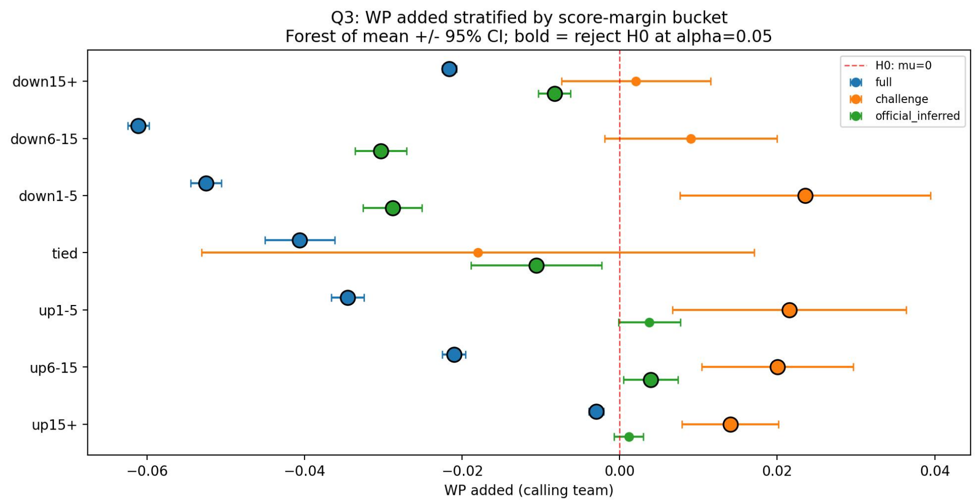

Figure 5: Q3 WP added stratified by calling-team score margin.

(−5.3 pp), but *halves* in Q4-clutch (−2.0 pp). This is the period where coaches are most strategic about timeout usage and where the WP model itself becomes most informative because the score margin grows in relative importance. TV-timeout WP cost likewise drops to zero in Q4-clutch (+0.0002 pp, ns).

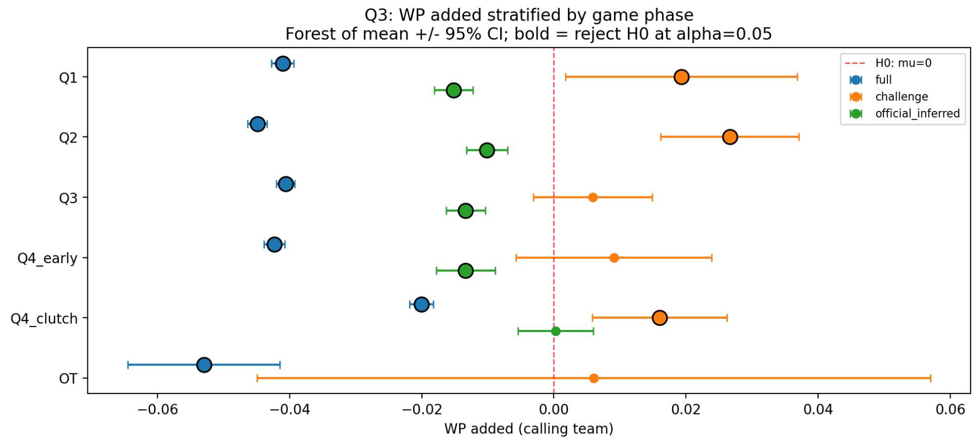

Figure 6: Q3 WP added stratified by game phase.

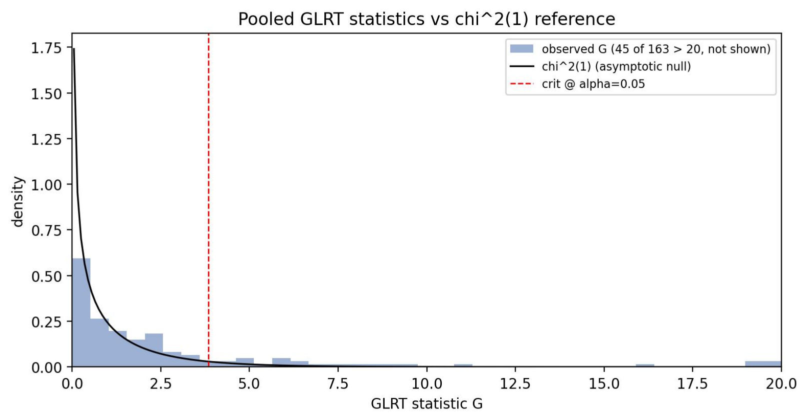

Figure 7: Pooled empirical distribution of GLRT statistics across all 163 stratified tests, overlaid on the asymptotic χ^2_1 null reference.

4.5 Pooled Distribution of Test Statistics

Across the four stratification schemes (overall, streak, margin, time) and the three impact metrics, we ran 163 stratified GLRTs and rejected H_0 in 69 (42.3%). The empirical distribution of G values is strongly right-shifted relative to its asymptotic χ^2_1 null (Figure 7), confirming that the rejections reflect real signal rather than the $\sim$5% false-positive rate one would expect under universal H_0.

5 Discussion

The three metrics behave differently because they absorb different sources of variance into the null. Q1 cancels by construction in the aggregate—the PPP-gap baseline is built precisely so that $\mathbb{E}[\bar{X}] = 0$ under no-effect—and only reveals signal once we stratify on a context that the baseline does not control for. Q2 leaves opponent variance in, which is why even the aggregate test rejects weakly. Q3 measures WP, which is non-linear in score margin near the endgame; consequently, the same $\sim$1-point swing produces drastically different WP movements depending on game state, and the GLRT exposes this with very high statistical power.

The signed-WP signal in Figure 2 is the clearest single result: *timeouts ride momentum, they don't reverse it.* Whether this reflects a genuine causal failure of the timeout (the calling team was on a hot streak and would have stayed hot anyway, but chose to call timeout) or selection bias (coaches call timeouts at very specific game states) cannot be answered with these between-group GLRTs alone. The companion "applied-causality" report on this same data resolves part of this ambiguity by adopting a within-game matched-twin design, and finds that conditioning on game identity flips the headline sign: matched-twin timeouts produce $+0.4$ pts of recovery rather than 0 or -1.3 pps of WP. The GLRT signal-detection view here is best understood as a precise measurement of the *observed* pattern; resolving *what causes it* requires the matching design, the substitution mediator analysis, or an instrument in the spirit of Weimer et al. (2023).

Limitations. (i) The GLRT relies on Gaussianity of the impact metric. Q1 and Q2 are noisy averages of point counts and are approximately Gaussian by CLT; Q3 is an output of a logistic model and is bounded in $[-1, 1]$, so the Gaussian assumption is violated near the endpoints. Empirically the t-test and GLRT p-values agree to four decimal places throughout, suggesting the violation is mild for n this large, but a non-parametric permutation test would be a robust check. (ii) The 163 tests are not independent (the same events appear in multiple stratifications) so a Bonferroni or Benjamini–Hochberg correction would be conservative; we report uncorrected p-values. (iii) The TV-timeout cohort is rulebook-inferred rather than labeled, with F1 ≈ 0.9 in the validation era, so the `official_inferred` effects should be read as a lower bound on the true mandatory-timeout effect.

Future work. The natural next step is to replace the unconditional null in Equation 1 with a counterfactual null that accounts for game state—e.g., comparing each timeout to a matched non-timeout possession in the same game, and applying the GLRT to the paired difference. This converts the GLRT into a within-pair causal estimator and inherits both the variance reduction of matching and the asymptotic distribution theory of the GLRT.

Acknowledgments and Disclosure of Funding

N/A

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